Huawei Access Node YEL-ACC-52 Standard Operating Procedure (SOP) - QoS Congestion Event Resolution

Node: YEL-ACC-52 | **Vendor:** Huawei | **Severity:** Minor | **Alarm Code:** ALARM_8040

Description

This SOP is designed to address the QoS congestion event, specifically microbursts and interface tail drops due to QoS queue starvation, on the Huawei access node YEL-ACC-52 at the Yelahanka Sub-station. The primary cause of this failure domain is attributed to the misconfiguration or inadequate tuning of Quality of Service (QoS) parameters, which can lead to excessive buffer occupancy and subsequent packet drops. Furthermore, the inherent characteristic of the Huawei's Enhanced Ethernet (EE) technology, which prioritizes packets based on their DSCP values, can exacerbate the issue if not properly configured. Additionally, the Huawei's Quality of Service (QoS) framework, which utilizes the concept of Traffic Policing and Traffic Shaping, can become ineffective if the policing and shaping rates are not accurately set, resulting in congestion and packet loss. It is crucial to carefully diagnose and address the root cause of the issue to prevent similar occurrences in the future.

Pre-requisite Checks

- check: Verify the status of the redundant link on the YEL-ACC-52 access node to ensure it is operational and not contributing to the congestion issue.

- check: Confirm the presence of a Huawei technician at the Yelahanka Sub-station to facilitate hands-on troubleshooting and resolution.

- check: Ensure that all relevant Huawei documentation and configuration files are readily available for reference.

Potential Root Causes

Cause	Probability
Misconfigured QoS parameters (e.g., policing rates, shaping rates, queue depths)	High
Insufficient buffer allocation for EE traffic	Medium
Inadequate traffic policing and shaping configurations	Medium

Diagnostic Commands & Expected Outputs

```
Command: display qos traffic-statistics, Expected Output: Traffic statistics for each QoS class will be displayed, including packet drop counts, buffer occupancy, and policing/shaping rates., Description: This command will provide valuable insights into the current QoS traffic statistics, helping to identify potential sources of congestion and packet drops.
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```
Command: display ee traffic-statistics, Expected Output: Enhanced Ethernet traffic statistics will be displayed, including packet drop counts, buffer occupancy, and EE-specific metrics., Description: This command will focus on the EE traffic statistics, allowing us to assess the performance of the Enhanced Ethernet technology and identify potential bottlenecks.

Command: display qos interface-statistics, Expected Output: QoS interface statistics will be displayed, including packet drop counts, buffer occupancy, and policing/shaping rates for each interface., Description: This command will provide detailed QoS interface statistics, enabling us to pinpoint specific interfaces that may be contributing to the congestion issue.
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Resolution Steps

Step 1: Step: Adjust QoS parameters (policing rates, shaping rates, queue depths), Command: set qos policing-rate and set qos shaping-rate , Description: Adjust the policing and shaping rates on the affected interfaces to mitigate congestion and packet drops. Ensure that the new rates are properly configured and verified before proceeding.

Step 2: Step: Increase buffer allocation for EE traffic, Command: set ee buffer-size , Description: Increase the buffer size for EE traffic on the affected interfaces to provide adequate space for packet buffering and reduce the likelihood of packet drops.

Step 3: Step: Re-configure traffic policing and shaping, Command: set qos policing-rate and set qos shaping-rate , Description: Re-configure the traffic policing and shaping on the affected interfaces to ensure accurate policing and shaping rates, preventing congestion and packet drops.

Post-Check Verification

```
Command: display qos traffic-statistics, Expected Output: Traffic statistics for each QoS class will be displayed, indicating improved packet delivery and reduced congestion., Description: Verify that the QoS traffic statistics show improved packet delivery and reduced congestion after the resolution steps.

Command: display ee traffic-statistics, Expected Output: Enhanced Ethernet traffic statistics will be displayed, indicating improved packet delivery and reduced congestion., Description: Verify that the EE traffic statistics show improved packet delivery and reduced congestion after the resolution steps.

Command: display qos interface-statistics, Expected Output: QoS interface statistics will be displayed, indicating improved packet delivery and reduced congestion., Description: Verify that the QoS interface statistics show improved packet delivery and reduced congestion after the resolution steps.
```

Roll-back Procedures

- Step: Restore original QoS parameter settings, Command: undo set qos policing-rate and undo set qos shaping-rate , Description: Restore the original QoS parameter settings to their pre-resolution values to mitigate the impact of the resolution steps.

- Step: Restore original EE buffer allocation, Command: undo set ee buffer-size , Description: Restore the original EE buffer allocation to its pre-resolution value to mitigate the impact of the resolution steps.

- Step: Restore original traffic policing and shaping configurations, Command: undo set qos policing-rate and undo set qos shaping-rate , Description: Restore the original traffic policing and shaping configurations to their pre-resolution values to mitigate the impact of the resolution steps.